

Visitor Pass Monitoring System

ASP.NET Core MVC Laboratory Exercise Specification

Project Objective

Develop an ASP.NET Core MVC application for security personnel to monitor visitors entering and leaving a company using Model Binding, Cookie Authentication, Data Annotation Validation, Repository Pattern (in-memory), Bootstrap, and Razor Views.

Business Scenario

ABC Corporate Office replaces its paper visitor logbook with a web application. Security personnel register visitors, monitor active visitors, and record exit times. Only authenticated users may access the application.

Learning Objectives

MVC Architecture, Model Binding, DTOs, Validation, Cookie Authentication, Authorization, Repository Pattern, CRUD, Search, Bootstrap.

Functional Requirements

User Registration, Login, Logout, Register Visitor, View Visitor Monitoring List, Edit Visitor, View Details, Record Visitor Exit, Search.

Visitor Fields

Pass Number (auto-generated), First Name, Last Name, Company, Contact Number, Person to Visit, Department, Purpose, Entry Date & Time, Valid ID Presented, Notes.

Monitoring

Display Pass Number, Visitor, Company, Person to Visit, Entry Time, Exit Time, Status, Actions.

Status

Inside Building, Left Building.

Validation

Use Required, StringLength, Phone, Display, DataType and other Data Annotation attributes. Validate ModelState.

Authentication

Public: Login, Register. Protected: Visitor pages using Cookie Authentication and [Authorize].

Entities

User: Id, FirstName, LastName, Email, Username, Password. Visitor: Id, PassNumber, FirstName, LastName, Company, ContactNumber, PersonToVisit, Department, Purpose, EntryDateTime, ExitDateTime, Status, ValidIdPresented, Notes.

Repositories

UserRepository and VisitorRepository using static List.

Controllers & Views

Controllers: HomeController, AccountController, VisitorController. Views: Login, Register, Index, Create, Edit, Details, Exit.

Submission

Complete MVC solution with authentication, CRUD, search, validation, Bootstrap UI, repository pattern, and clean code.